Review for First Semester Exam Grade 12 (Social Science and Science)

A. Multiple choice questions

1. The root mean square speed of the molecules of an enclosed gas is v. What will be the mean square speed if the pressure is doubled, and the temperature remains the same?

a. $v/2$ **b.** v

c. $2v$ **d.** $4v$

2. If the volume of an ideal gas is doubled while its temperature is quadrupled, does the pressure

a. remains the same **b.** decrease by a factor of 2

c. decrease by a factor of 4 **d.** increase by a factor of 2

3. A 2.0 moles of an ideal gas system are maintained at a constant volume of 4 liters. If 100 J of thermal energy is transferred to the system, what is the change in the internal energy of the system?

a. 0 J **b.** 400 J

c. 70 J **d.** 100 J

4. A container holds 0.50 m^3 of oxygen at an absolute pressure of 4.0 atm. A valve is opened, allowing the gas to drive a piston, increasing the volume of the gas until the pressure drops to 1.0 atm. If the temperature remains constant, what new volume does the gas occupy?

a. 1.0 m^3 **b.** 1.5 m^3

c. 2.0 m^3 **d.** 0.12 m^3 **e.** 2.5 m^3

5. An ideal gas is maintained at a constant pressure of 70.0 kPa during an isobaric process while its volume decreases by 0.20 m^3. What is the work done by the system on its environment?

a. 14 kJ **b.** 35 kJ

c. -14 kJ **d.** 235 kJ

6. What is the internal energy of 26.0 g of neon gas at a temperature of 152°C?

a. 2 440 J **b.** 6 830 J

c. 3 140 J **d.** 5 870 J **e.** 5 020 J

7. If the average kinetic energy of molecules is higher, then temperature of gas is

a. high **b.** low

c. zero **d.** infinite

8. Which of the assumptions below is *not* made in the kinetic theory of gases?

a. The number of molecules is very small.
b. The molecules obey Newton's laws of motion.
c. The collisions between molecules are elastic.
d. The gas is a pure substance.
e. The average separation between molecules is large compared with their dimensions.

9. Which of the following formula is correct for universal gas constant?

a. $R = \frac{PV}{nT}$ **b.** $R = \frac{V}{PT}$

c. $R = \frac{nT}{PV}$ **d.** None of these above

10. If the temperature of an ideal gas is increased from 200 K to 600 K, what happens to the rms speed of the molecules?

a. It increases by a factor of 3.
b. It remains the same.
c. It is one-third of the original speed.
d. It is $\sqrt{3}$ times the original speed.
e. It increases by a factor of 6.

11. A monatomic ideal gas expands from 1.00 m^3 to 2.50 m^3 at a constant pressure of 2.00×10^5 Pa. Find the change in the internal energy of the gas.

a. 7.50×10^5 J **b.** 1.05×10^6 J

c. 4.50×10^5 J **d.** 3.00×10^5 J

e. -4.50×10^5 J

12. An ideal gas drives a piston as it expands from 1.00 m^3 to 2.00 m^3 at a constant temperature of 850 K. If there are 390 moles of gas in the piston, how much work does the gas do in displacing the piston?

a. 1.9×10^6 J **b.** 2.5×10^6 J

c. 4.7×10^6 J **d.** 2.1×10^5 J

e. 3.5×10^6 J

13. An engine does 15 kJ of work while rejecting 37 kJ to the cold reservoir. What is the efficiency of the engine?

a. 0.15 **b.** 0.29

c. 0.33 **d.** 0.45 **e.** 1.2

14. A steam turbine operates at a boiler temperature of 450 K and an exhaust temperature of 3.0×10^2 K. What is the maximum theoretical efficiency of this system?

a. 0.24 **b.** 0.50

c. 0.33 **d.** 0.67 **e.** 0.15

15. When an ideal gas undergoes an adiabatic expansion, which of the following statements is true?

a. The temperature of the gas doesn't change.
b. No work is done by the gas.
c. No energy is delivered to the gas by heat.
d. The internal energy of the gas doesn't change.
e. The pressure increases.

16. If an ideal gas undergoes an isobaric process, which of the following statements is true?
a. The temperature of the gas doesn't change.
b. Work is done on or by the gas.
c. No energy is transferred by heat to or from the gas.
d. The volume of the gas remains the same.
e. The pressure of the gas decreases uniformly.

17. An ideal gas is compressed to half its initial volume by means of several possible processes. Which of the following processes result in the most work done on the gas?
a. isothermal **b.** adiabatic **c.** isobaric
d. The work done is independent of the process.

18. How much net-work is done by the gas undergoing the cyclic process illustrated in Figure MCQ12.4?
Choose the best estimate.
a. 1×10^5 J **b.** 2×10^5 J **c.** 3×10^5 J
d. 4×10^5 J **e.** 5×10^5 Pa

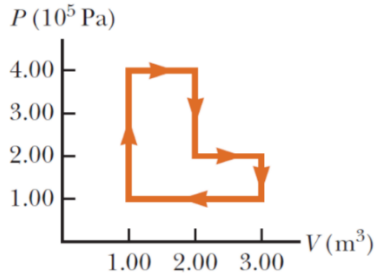


Figure MCQ12.4

19. The distance between the crest of a water wave and the next trough is 2 m. If the frequency of a particular wave is 2 Hz, what is the speed of the wave?
a. 4 m/s **b.** 1 m/s **c.** 8 m/s **d.** 2 m/s
e. impossible to determine from the information given

20. The position of an object moving with simple harmonic motion is given by $x = 4\cos(6\pi t)$, where x is in meters and t is in seconds. What is the period of the oscillating system?
a. 4 s **b.** 16 s **c.** 13 s **d.** 6π s
e. impossible to determine from the information given

21. A sound wave traveling in air has a frequency f and wavelength λ. A second sound wave traveling in air has wavelength $\lambda/2$. What is the frequency of the second sound wave?
a. $4f$ **b.** $2f$ **c.** f **d.** $1/2f$
e. $1/4\ f$

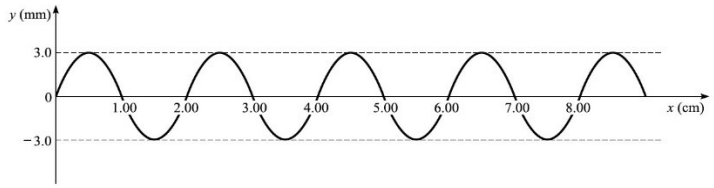
22. The position of the nodes in a standing wave is given by:
a. $x = n\frac{\lambda}{4}$ **b.** $x = n\frac{\lambda}{2}$ **c.** $x = (2n+1)\frac{\lambda}{4}$ **d.** $x = \frac{\lambda}{2}$

23. The position of the antinodes in a standing wave is given by:
a. $x = n'\frac{\lambda}{4}$ **b.** $x = n\frac{\lambda}{2}$ **c.** $x = (2n+1)\frac{\lambda}{2}$ **d.** $x = \frac{\lambda}{2}$

24. Sound waves of wavelength λ traveling with velocity v in a medium enter another medium in which their velocity becomes $4v$. What is the wavelength of the sound waves in the second medium?
a. $\lambda/4$ **b.** λ **c.** 2λ **d.** 4λ
e. 16λ

25. The velocity of sound in sea water is 1530 m/s. If a sound of frequency 1800 Hz produced in sea water, what is its wavelength?
a. 0.85 m **b.** 1.15 m **c.** 0.50 m **d.** 1.30 m
e. 2.00 m

26. The wave shown in Fig is being sent out by a 60 cycle/s vibrator. Find the wavelength and wave speed.



a. $\lambda = 2$ cm, $v = 1.2$ cm/s **b.** $\lambda = 2$ cm, $v = 120$ cm/s
c. $\lambda = 2$ m, $v = 1.2$ m/s **d.** $\lambda = 4.5$ cm, $v = 1.2$ m/s

B. Fill in the blanks

27. The manner in which a state of a system can change from an initial state to a final state is called .....
28. ..... is a system in which both matter and energy do not interact with the outside environment.
29. A thermodynamic process in which the pressure of the system does not change with time is called.....
30. The ..... is the process in which the temperature of a system remains constant.
31. A ..... is a thermodynamic process in which a system returns to its initial state and $\Delta U = 0$.
32. The total kinetic energy of all molecules in a monatomic ideal gas is proportional to .....
33. The root-mean-square speed of gas molecules is given by .....
34. A ..... in a standing wave is a point where displacement is always zero.
35. An ..... in a standing wave is a point where the amplitude is maximum.
36. The distance between two consecutive nodes in a standing wave is .....
37. The distance between a node and the nearest antinode is .....
38. The positions of nodes along a string of wavelength λ are given by .....
39. The positions of antinodes along a string of wavelength λ are given by .....

C. Critical thinking questions

40. State the kinetic theory of ideal gases.
41. What does the first law of thermodynamics study?
42. What is a system in thermodynamics?
43. What is a thermodynamic system?
44. What is the state of a system in thermodynamics?
45. What is the difference between a cyclic process and a non-cyclic process?
46. What is the internal energy of an ideal gas?
47. Describe isochoric, isobaric, and isothermal processes.
48. State the principle initial and final states of a system.
49. State the principle of equivalence in a closed system.
50. What is an adiabatic process?
51. Describe the processes involved in Carnot's cycle.
52. What is the difference between an internal combustion engine and an external combustion engine? Provide one example of each.
53. State Carnot's theorem and provide its formula.
54. Describe the four processes involved in a four-stroke heat engine.
55. Describe the two processes involved in a two-stroke heat engine.
56. What is a wave?
57. What is the superposition of waves?
58. What is the amplitude of a wave?
59. What is the phase of a wave?
60. What is a standing wave?
61. What is the phenomenon of resonance?
62. How many types of resonance exist? Define each type.
63. Is the amplitude of a superposition wave the same as that of individual oscillations? Explain.
64. What is sinusoidal motion?
65. What is the maximum displacement when two sinusoidal oscillatory motions are in antiphase?
66. Why are compass boxes made of copper or plastic instead of iron?
67. How is the right-hand rule used to determine the direction of the magnetic field in a straight current-carrying wire or circular loops?
68. What is a magnet? How many types of magnets are there? Describe each type and provide examples.
69. What are magnetic sources? What is the SI unit of magnetic induction?
70. When an electric current passes through a solenoid, magnetic field lines are observed to emerge from one pole and loop back to another. Which pole is the north pole, and which is the south pole?
71. How does the magnetic field at the center of a solenoid change when the electric current is doubled? What happens if the direction of the current is reversed?
72. What is a magnetic field?
73. What is a uniform magnetic field? Where can such a field be located?

D. Work problems

74. At what temperature (in Degree Kelvin) will the oxygen molecules have the same root mean square speed (v_{rms}) as hydrogen molecules at temperature of 200 K?
75. A cylinder contains a mixture of helium and argon gas in equilibrium at 150°C.
(a) What is the average kinetic energy for each type of gas molecule?
(b) What is the rms speed of each type of molecule?
76. In an ultrahigh vacuum system (with typical pressures lower than 10^{-7} pascal), the pressure is measured to be 1.00×10^{-10} torr (where 1 torr = 133 Pa). Assuming the temperature is 300 K, find the number of molecules in a volume of 1.00 m^3.
77. What is the average kinetic energy of a molecule of oxygen at a temperature of 300 K?
78. An 8.31 liters vessel contains 2.00 moles of an ideal gas at a pressure of 1.50×10^6 Pa. Find the temperature of the ideal gas. Given that $R = 8.31 J/mol.K$
79. A gas is compressed at a constant pressure of 0.80 atm from 9.00L to 2.00 L. In the process, 400 J of energy leaves the gas by heat.
a. What is the work done on the gas?
b. What is the change in its internal energy?
80. A spherical balloon of volume 4.00×10^3 cm^3 contains helium at a pressure of 1.20×10^5 Pa. How many moles of helium are in the balloon if the average kinetic energy of each helium atom is 3.60×10^{-22} J?
81. In each cycle of its operation, a heat engine expels 2400 J of energy and performs 1800 J of mechanical work.
a. How much thermal energy must be added to the engine in each cycle?
b. Find the thermal efficiency of the engine.
82. (a) How many atoms of helium gas fill a spherical balloon of diameter 30.0 cm at 20.0°C and 1.00 atm?
(b) What is the average kinetic energy of the helium atoms?
(c) What is the rms speed of the helium atoms?
83. A wave is described by $y = 100 \sin(100\pi t + \pi/4)$ cm. Find the amplitude, period, Frequency, and phase constant of the wave.
84. Fifteen identical particles have various speeds: one has a speed of 2.00 m/s, two have speeds of 3.00 m/s, three have speeds of 5.00 m/s, four have speeds of 7.00 m/s, three have speeds of 9.00 m/s, and two have speeds of 12.0 m/s. Find
(a) the average speed,
(b) the rms speed.

85. A container with 1 m^3 of helium gas is at standard temperature and pressure (STP).

(a) Determine the density of helium, given that the molar mass of helium is 4 g/mol.

(b) Calculate the mass of one helium molecule.

(c) Find the root mean square (rms) speed and average kinetic energy of the gas molecules.

86. A container of hydrogen gas has a volume of 8.01 L at -10°C and a pressure of 0.5 MPa. The molar mass of hydrogen is 2 g/mol:

(a) Calculate the mass of hydrogen in the container.

(b) Determine the root mean square (rms) speed of the hydrogen molecules.

(c) Calculate the average kinetic energy per molecule and the total kinetic energy of the gas.

87. A cube filled with helium gas has an edge length of 30.0 cm. The gas is at 20.0°C and 1.00 atm:

(a) How many molecules of helium are in the cube?

(b) Calculate the average kinetic energy of the helium molecules.

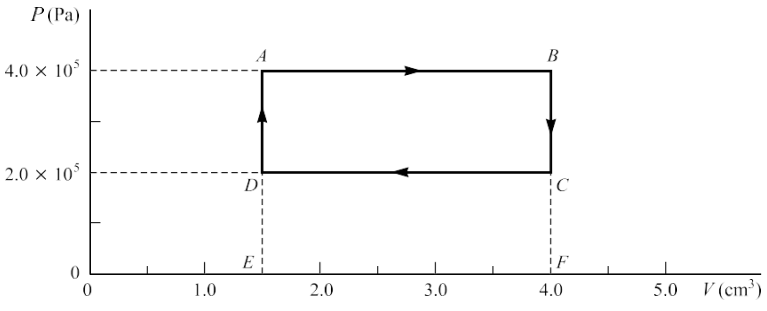
(c) Find the root mean square (rms) speed of helium molecules.

88. An ideal gas undergoes an isothermal compression from an initial volume of 0.05 m^3 to a final volume of 0.02 m^3 under a final pressure of 2 atm:

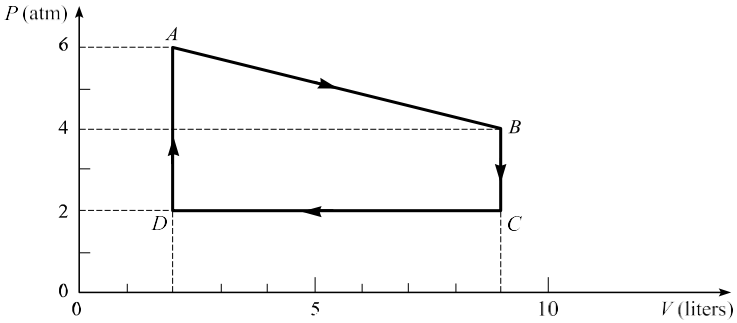
(a) Determine the initial pressure and the ratio of the initial volume to the final volume ($V_i \rightarrow V_f$).

(b) Does the gas absorb or release heat during this process?

89. The P-V diagram in Fig. 20-1 applies to a gas undergoing a cyclic change in a piston–cylinder arrangement. What is the work done by the gas in **(a)** portion AB of the cycle? **(b)** Portion BC? **(c)** Portion CD? **(d)** Portion DA?



90. Find the network output per cycle for the thermodynamic cycle in Fig. 20-4. Give your answer to two significant figures. Ans. 2.1 kJ



91. The internal energy of an ideal gas changes by 415.5 J when its temperature increases from 2°C to 22°C in an isobaric process:

(a) Calculate the number of moles of the gas.

(b) Determine the work done by the gas and the heat absorbed during the process.

92. 50 J of thermal energy was removed from a gas and its internal energy decreased by 10 J.

(a) How much work was done?

(b) Was the work done on the gas, or by the gas?

93. The internal energy of 3.2 moles of some helium gas, at an initial temperature of 302 K, rose by 540 J.

(a) What was the final temperature of the gas?

(b) If, at the same time, the gas expanded from a volume of 220 cm^3 to 380 cm^3 at constant pressure of 2.3 × 10^5 Pa, how much work was done?

(c) How much thermal energy was transferred?

94. One mole of gas initially at a pressure of 2.00 atm and a volume of 0.300 L has an internal energy equal to 91.0 J. In its final state, the gas is at a pressure of 1.50 atm and a volume of 0.800 L, and its internal energy equals 182 J. For the paths IAF, IBF, and IF in **Figure P12.22**, calculate

(a) the work done on the gas and

(b) the net energy transferred to the gas by heat in the process.

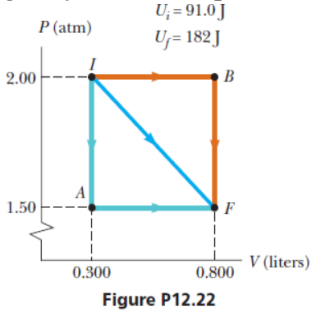


Figure P12.22

95. A steam engine operating between a boiler temperature of 220 °C and a condenser temperature of 35.0 °C delivers 8.00 hp. If its efficiency is 30.0 percent of that for a Carnot engine operating between these temperature limits, how many calories are absorbed each second by the boiler? How many calories are exhausted to the condenser each second?